Multilayer local score with an entropy-barrier window

Does a second, larger-window LS layer raise Inception Score?

Sampler step count is the first knob that improves quality without memorizing more: the champion window schedule resampled onto 60 steps takes CIFAR-10 FID 83.3 to 78.8 and IS 6.44 to 6.75, while six shape controls only trade one for the other.

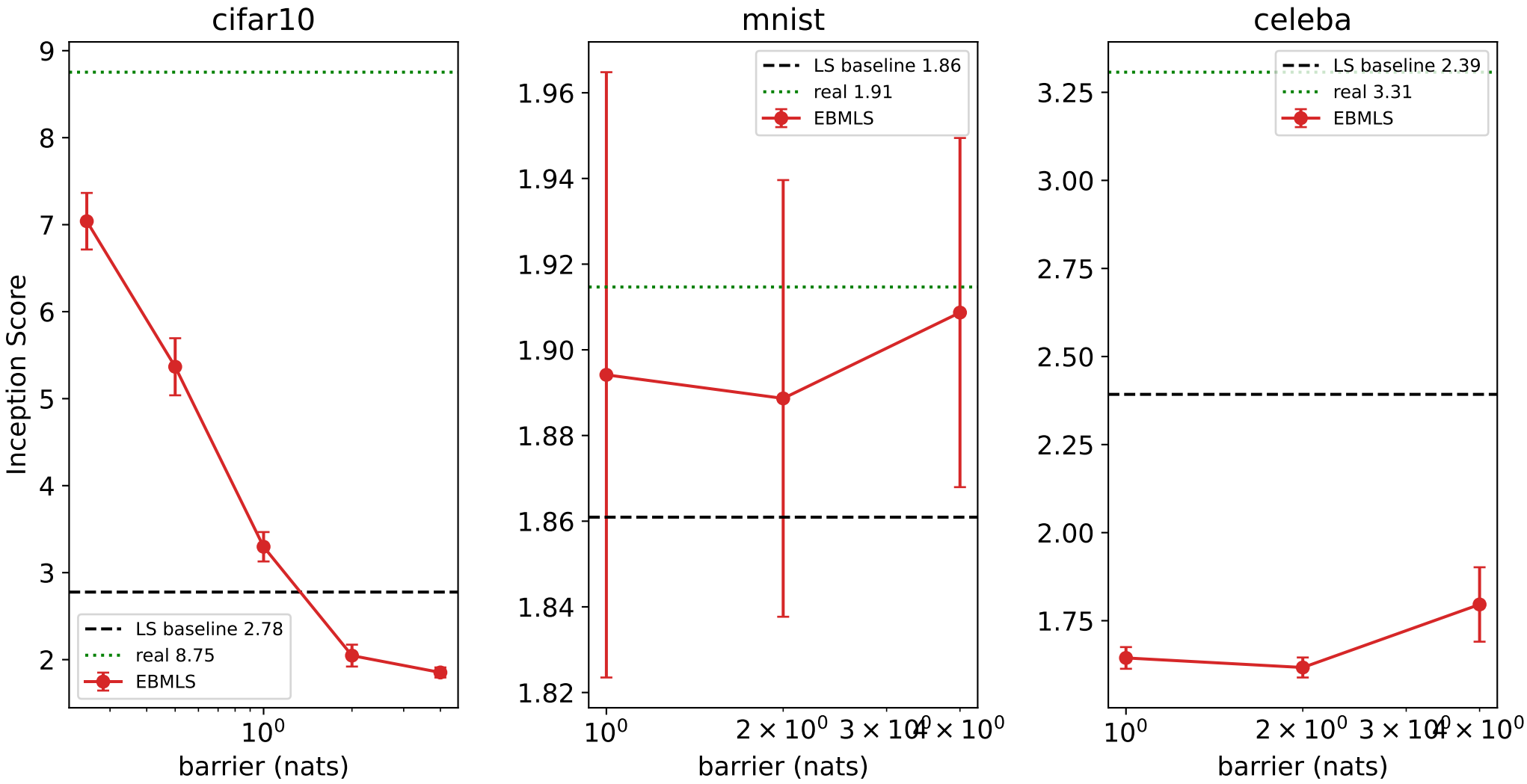
Layer 2 sees only layer 1's estimate, forms a posterior over all training images in a larger window, and adds the expected error of that estimate. The window at each step is the largest whose posterior entropy is still above a barrier — i.e. has not collapsed onto one image.

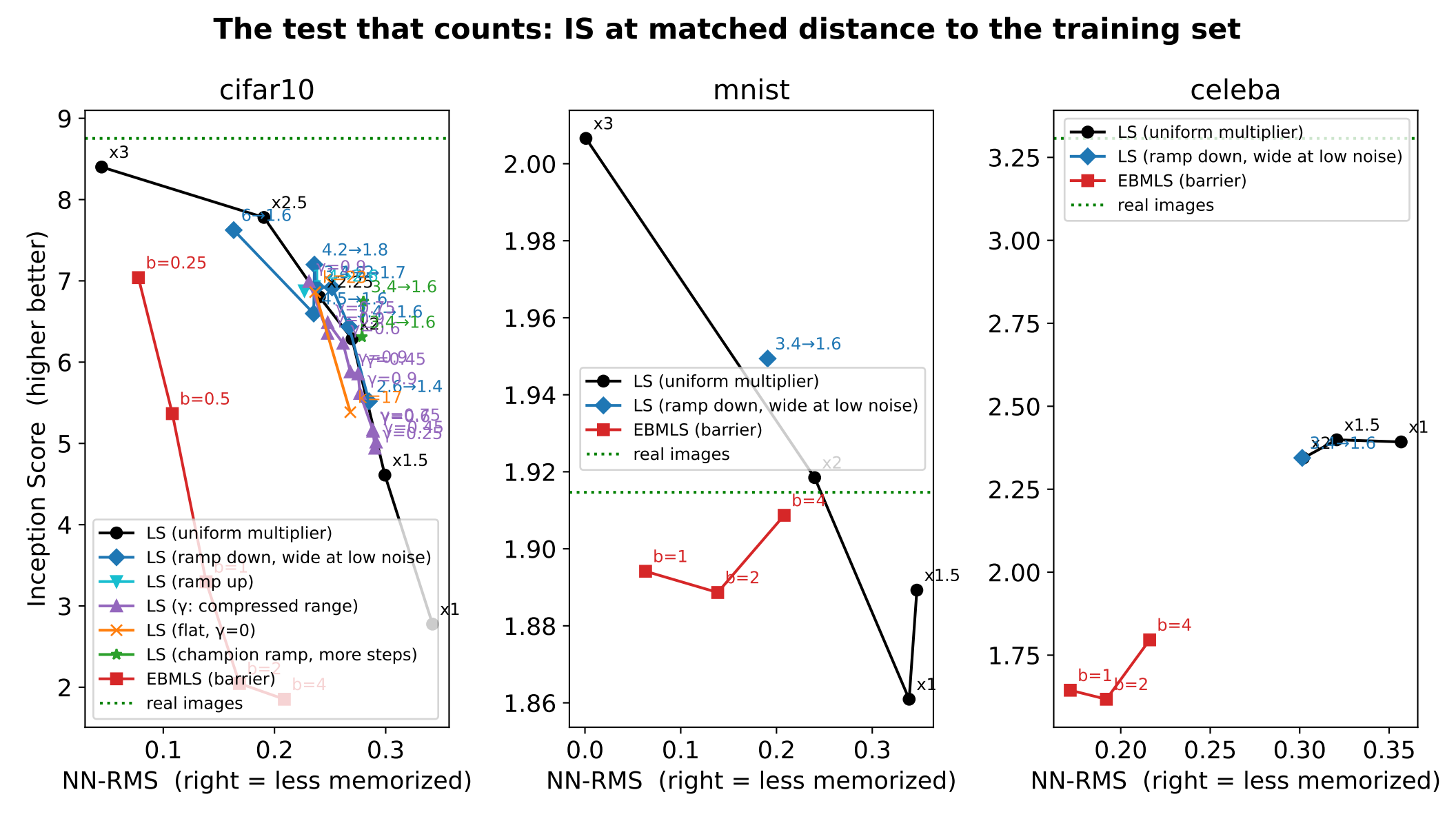
2026-10-22 · commit f5f1733 · seed 0 · 500 samples/config · IS over 5 splits · FID vs the full training set

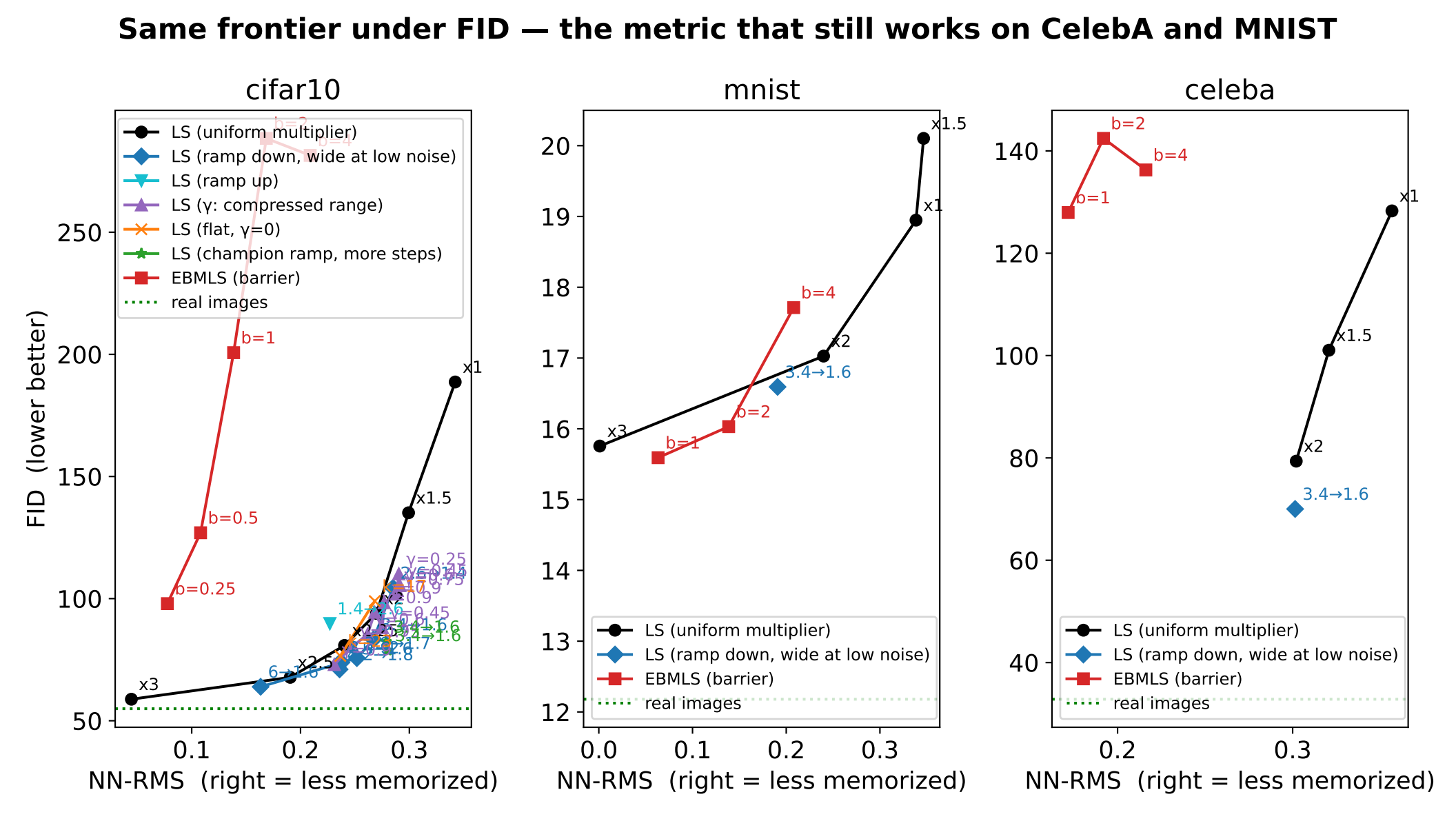
Metrics

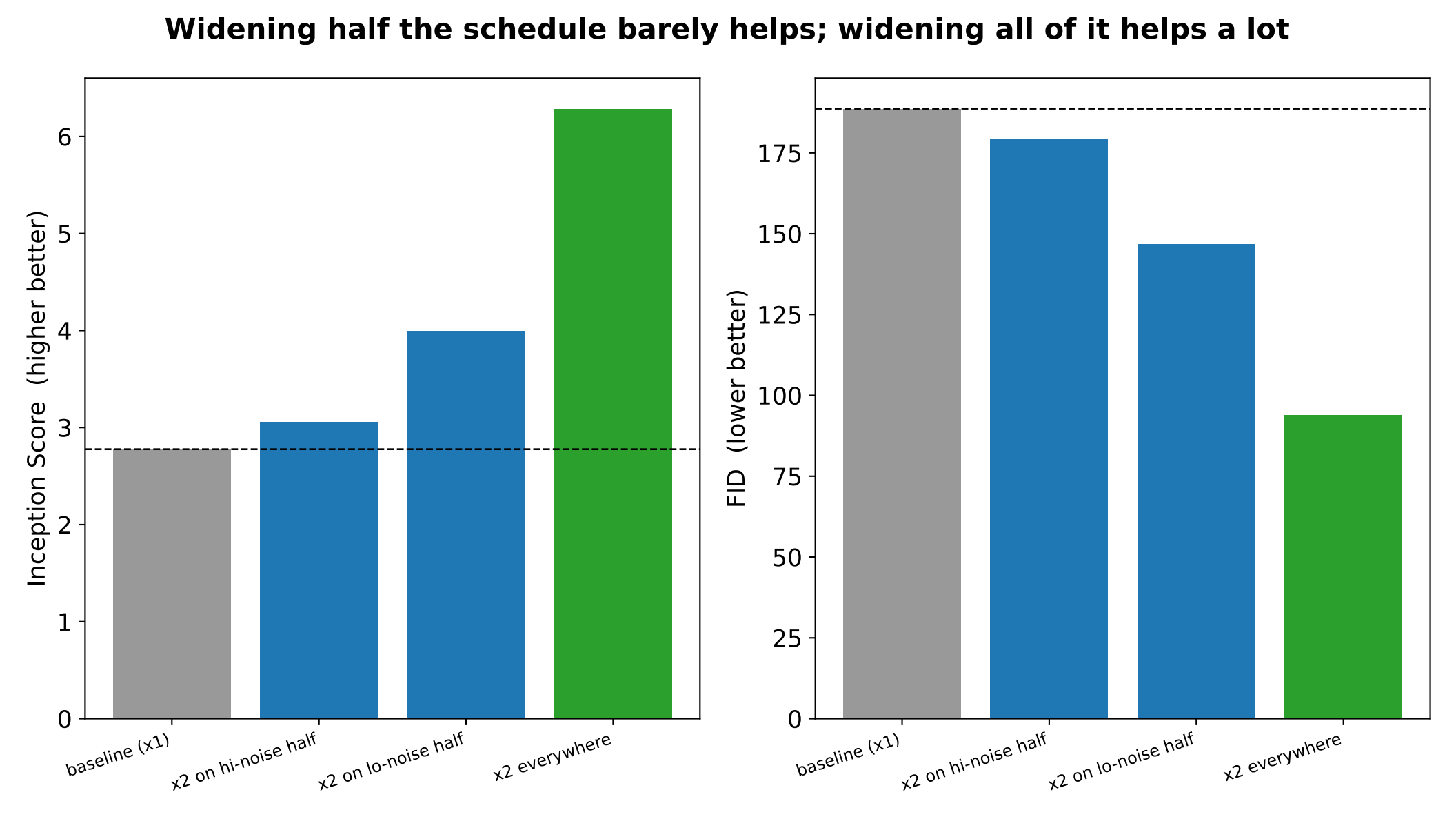
IS higher is better, FID lower is better; NN-RMS higher = less memorized

dataset	config	seed	IS	FID	NN-RMS	copy frac
celeba	real images	0	3.307 ± 0.119	32.8	—	—
celeba	LS	0	2.393 ± 0.043	128.3	0.357	0.000
celeba	LS (windows x1.5)	0	2.399 ± 0.214	101.0	0.321	0.000
celeba	LS (windows x2)	0	2.344 ± 0.090	79.4	0.302	0.000
celeba	LS (ramp x3.4→x1.6)	0	2.344 ± 0.193	70.0	0.301	0.000
celeba	EBMLS barrier 1	0	1.644 ± 0.031	127.9	0.172	0.000
celeba	EBMLS barrier 2	0	1.617 ± 0.028	142.4	0.192	0.000
celeba	EBMLS barrier 4	0	1.796 ± 0.106	136.3	0.216	0.000
cifar10	real images	0	8.753 ± 0.465	54.9	—	—
cifar10	LS	0	2.777 ± 0.157	188.7	0.342	0.000
cifar10	LS	1	2.857 ± 0.061	190.4	0.343	0.000
cifar10	LS (windows x1.5)	0	4.610 ± 0.215	135.1	0.299	0.000
cifar10	LS (windows x2)	0	6.286 ± 0.563	94.0	0.270	0.000
cifar10	LS (windows x2)	1	6.073 ± 0.590	93.0	0.274	0.000
cifar10	LS (windows x2, hi-noise half)	0	3.061 ± 0.179	179.2	0.286	0.000
cifar10	LS (windows x2, lo-noise half)	0	3.997 ± 0.252	146.8	0.299	0.000
cifar10	LS (windows x2.25)	0	6.803 ± 0.365	80.8	0.240	0.028
cifar10	LS (windows x2.5)	0	7.780 ± 0.360	67.8	0.191	0.146
cifar10	LS (windows x3)	0	8.401 ± 0.467	58.8	0.045	0.514
cifar10	LS (ramp x2.6→x1.4)	0	5.519 ± 0.456	104.4	0.285	0.000
cifar10	LS (ramp x3.4→x2)	0	6.925 ± 0.493	73.9	0.238	0.042
cifar10	LS (ramp x3.4→x1.6)	0	6.437 ± 0.310	83.3	0.267	0.002
cifar10	LS (ramp x3.4→x1.6)	1	6.314 ± 0.284	86.1	0.271	0.000
cifar10	LS (ramp x3.8→x1.7)	0	6.919 ± 0.417	75.6	0.252	0.034
cifar10	LS (ramp x4.2→x1.8)	0	7.200 ± 0.573	71.0	0.236	0.092
cifar10	LS (ramp x4.5→x1.6)	0	6.598 ± 0.433	73.4	0.235	0.054
cifar10	LS (ramp x6→x1.6)	0	7.625 ± 0.540	63.9	0.163	0.310
cifar10	LS (ramp x1.4→x2.6)	0	6.877 ± 0.297	89.8	0.227	0.016
cifar10	LS (γ=0.25 at x2)	0	4.941 ± 0.322	110.1	0.290	0.000
cifar10	LS (γ=0.45 at x2)	0	5.016 ± 0.158	105.3	0.291	0.000
cifar10	LS (γ=0.6 at x2)	0	5.151 ± 0.325	103.6	0.289	0.000
cifar10	LS (γ=0.75 at x2)	0	5.167 ± 0.409	101.8	0.288	0.000
cifar10	LS (γ=0.9 at x2.05)	0	5.615 ± 0.222	98.3	0.277	0.000
cifar10	LS (γ=0.9 at x2.2)	0	5.880 ± 0.333	94.4	0.268	0.000
cifar10	LS (γ=0.9 at x2.35)	0	6.355 ± 0.311	80.6	0.248	0.020
cifar10	LS (γ=0.45 at x2.5)	0	5.855 ± 0.412	88.1	0.275	0.002
cifar10	LS (γ=0.6 at x2.5)	0	6.232 ± 0.326	85.5	0.262	0.004
cifar10	LS (γ=0.75 at x2.5)	0	6.488 ± 0.447	80.4	0.248	0.012
cifar10	LS (γ=0.9 at x2.5)	0	6.992 ± 0.202	73.0	0.231	0.058
cifar10	LS (flat k=17)	0	5.384 ± 0.342	99.0	0.268	0.000
cifar10	LS (flat k=23)	0	6.862 ± 0.503	76.5	0.237	0.044
cifar10	LS (ramp, 40 steps)	0	6.310 ± 0.267	82.3	0.279	0.010
cifar10	LS (ramp, 60 steps)	0	6.746 ± 0.339	78.8	0.280	0.016
cifar10	EBMLS barrier 0.25	0	7.040 ± 0.325	97.9	0.078	0.196
cifar10	EBMLS barrier 0.5	0	5.367 ± 0.328	127.0	0.108	0.058
cifar10	EBMLS barrier 1	0	3.298 ± 0.169	200.7	0.139	0.004
cifar10	EBMLS barrier 2	0	2.046 ± 0.126	288.4	0.169	0.000
cifar10	EBMLS barrier 4	0	1.853 ± 0.058	281.5	0.209	0.000
mnist	real images	0	1.915 ± 0.027	12.2	—	—
mnist	LS	0	1.861 ± 0.062	18.9	0.339	0.000
mnist	LS (windows x1.5)	0	1.889 ± 0.054	20.1	0.347	0.000
mnist	LS (windows x2)	0	1.918 ± 0.091	17.0	0.240	0.078
mnist	LS (windows x3)	0	2.007 ± 0.082	15.8	0.001	0.846
mnist	LS (ramp x3.4→x1.6)	0	1.949 ± 0.082	16.6	0.191	0.250
mnist	EBMLS barrier 1	0	1.894 ± 0.071	15.6	0.063	0.460
mnist	EBMLS barrier 2	0	1.889 ± 0.051	16.0	0.139	0.160
mnist	EBMLS barrier 4	0	1.909 ± 0.041	17.7	0.208	0.008

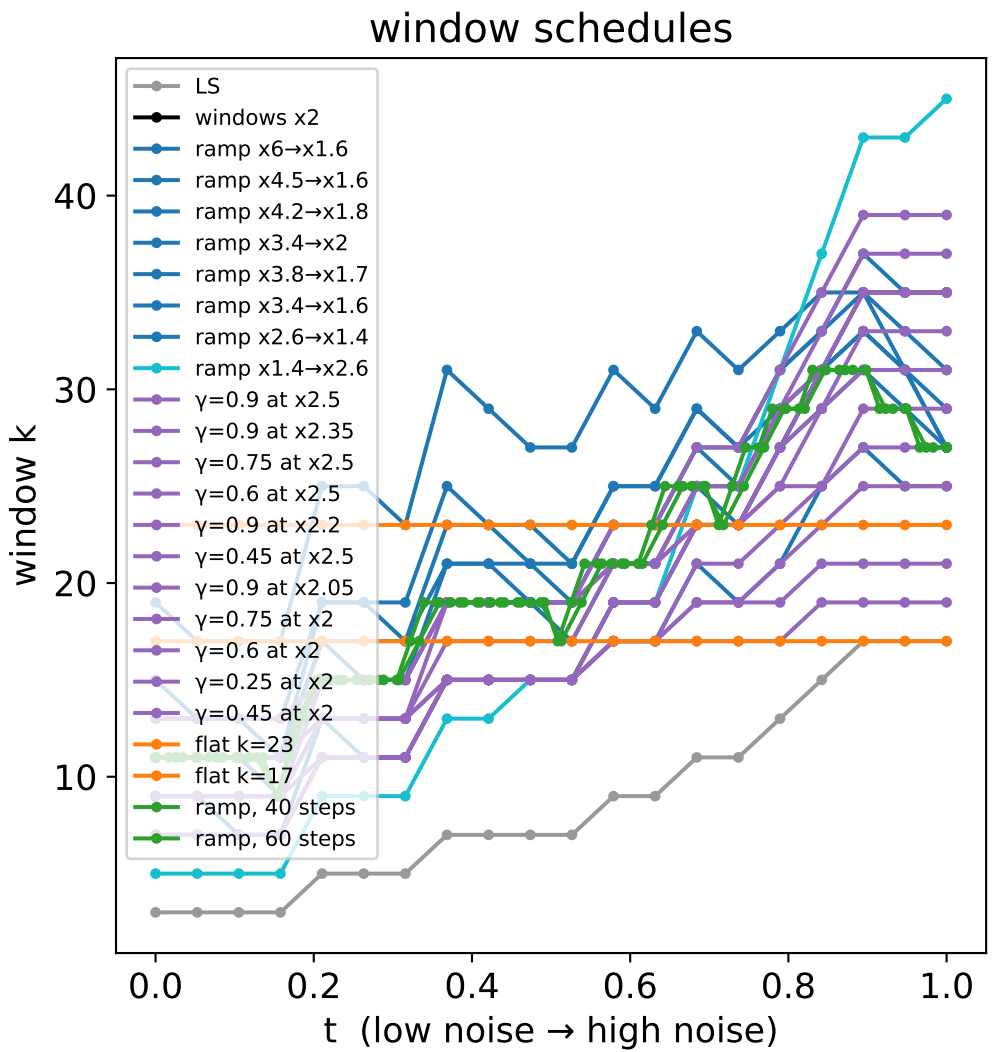

Inception Score vs entropy barrier
cifar10
mnist
celeba
Inception Score
barrier (nats)
LS baseline 2.78
real 8.75
EBMLS
LS baseline 1.86
real 1.91
EBMLS
LS baseline 2.39
real 3.31
EBMLS

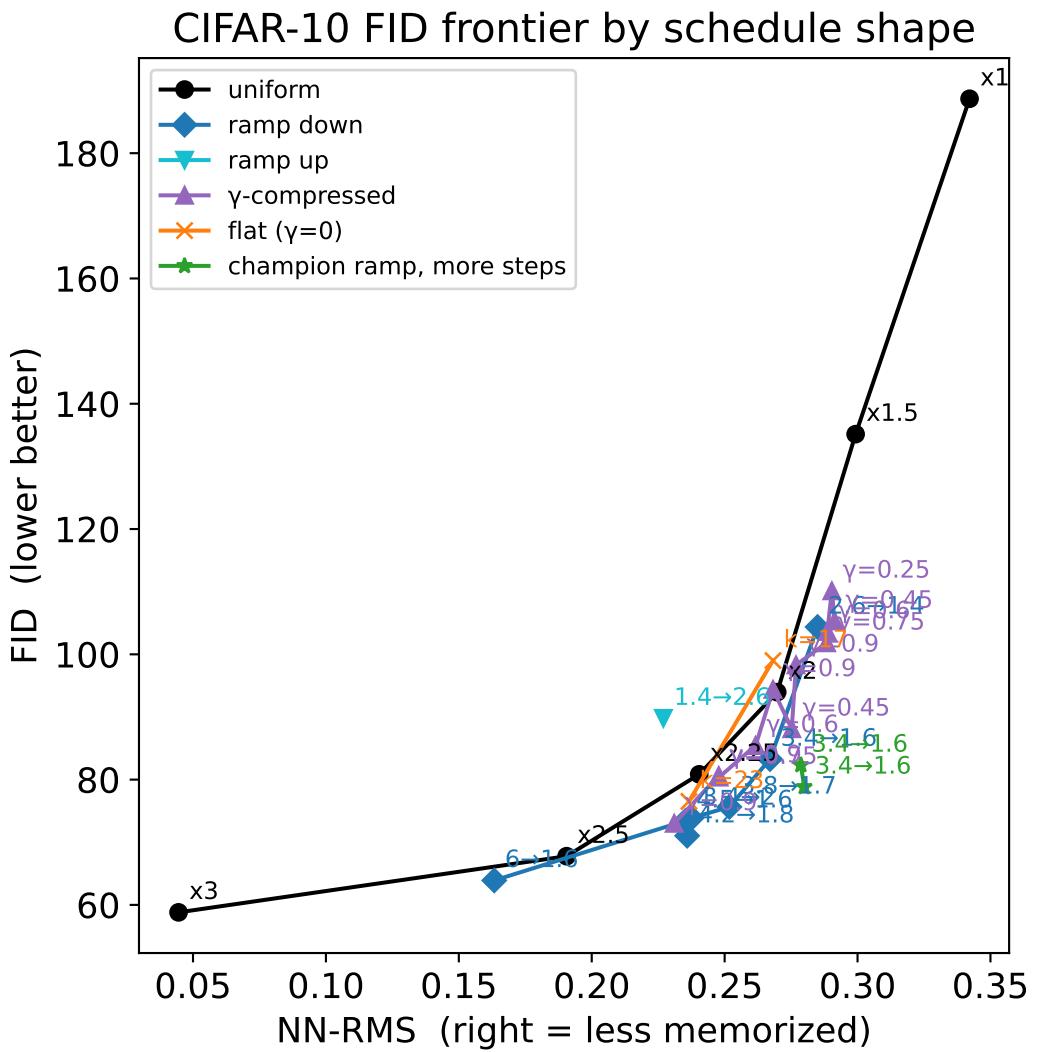

The test that counts: IS at matched distance to the training set
cifar10
mnist
celeba
Inception Score (higher better)
NN-RMS (right = less memorized)
LS (uniform multiplier)
LS (ramp down, wide at low noise)
LS (ramp up)
LS (γ: compressed range)
LS (flat, γ=0)
LS (champion ramp, more steps)
EBMLS (barrier)
real images
x3
x2.5
x2.25
x2
x1.5
x1
b=0.25
b=0.5
b=1
b=2
b=4
6→1.6
4.2→1.8
3.4→1.6
2.6→1.4
γ=0.9
γ=0.45
γ=0.25

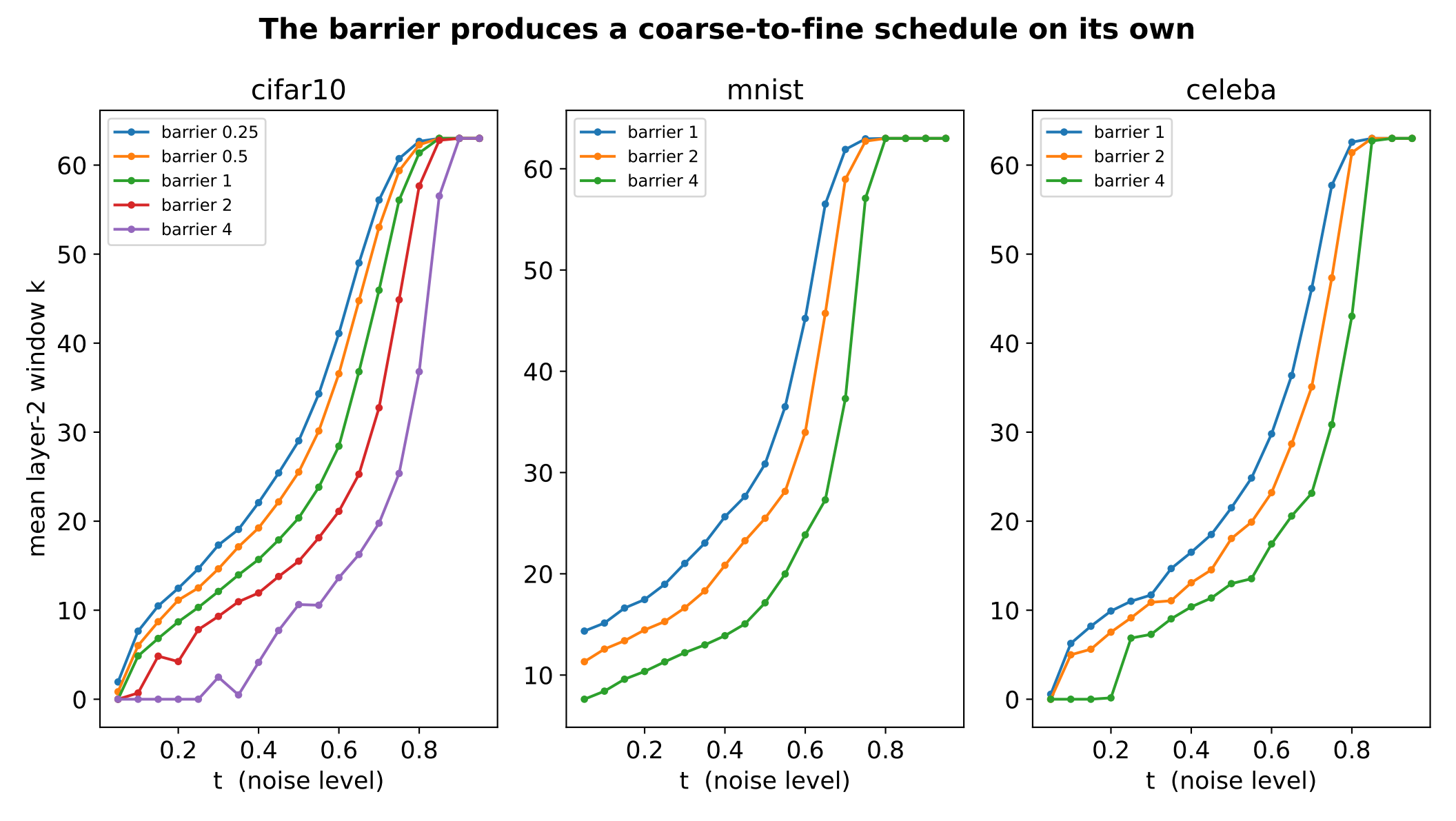

Same frontier under FID — the metric that still works on CelebA and MNIST
cifar10
mnist
celeba
FID (lower better)
NN-RMS (right = less memorized)
LS (uniform multiplier)
LS (ramp down, wide at low noise)
LS (ramp up)
LS (γ: compressed range)
LS (flat, γ=0)
LS (champion ramp, more steps)
EBMLS (barrier)
real images
x3
x2
x1.5
x1
b=1
b=2
b=4
b=0.5
b=0.25
3.4→1.6


Widening half the schedule barely helps; widening all of it helps a lot
Inception Score (higher better)
0
1
2
3
4
5
6
baseline (x1)
x2 on hi-noise half
x2 on lo-noise half
x2 everywhere
FID (lower better)
0
25
50
75
100
125
150
175
baseline (x1)
x2 on hi-noise half
x2 on lo-noise half
x2 everywhere

Tilting the schedule beats widening it uniformly

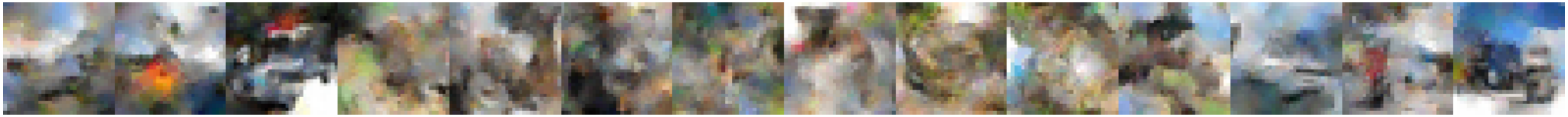





The barrier produces a coarse-to-fine schedule on its own
cifar10
mnist
celeba
barrier 0.25
barrier 0.5
barrier 1
barrier 2
barrier 4
mean layer-2 window k
t (noise level)
0
10
20
30
40
50
60
0.2
0.4
0.6
0.8

cifar10 samples: baseline, best non-copying, best EBMLS, and the memorizing end

LS — IS 2.78, NN-RMS 0.342, copies 0.0%



LS (ramp x3.4→x1.6) — IS 6.44, NN-RMS 0.267, copies 0.2%



EBMLS barrier 0.25 — IS 7.04, NN-RMS 0.078, copies 19.6%



LS (windows x3) — IS 8.40, NN-RMS 0.045, copies 51.4%



Where this leaves us (1/3)

1. The entropy-barrier multilayer layer does not pay off. Plain LS with wider windows dominates it at every matched memorization level, on both metrics. Layer 2's weights sum to 1, so it *replaces* layer 1's estimate rather than refining it. (hshunt is re-deriving the layer-2 posterior.)

2. The result came from the control: the per-step window schedule, not the layer count, moves the metrics. Uniform x2 takes CIFAR-10 FID 189 -> 94, CelebA 128 -> 79, MNIST 19.0 -> 17.0, with zero training-set copies. (IS and FID disagreed three times here; FID won each time.)

Where this leaves us (2/3)

3. And the schedule's *shape* beats its width. Ramping the multiplier down from 3.4 at t=0 to 1.6 at t=1 beats uniform x2 at matched memorization and zero copies -- CIFAR-10 FID 83/86 vs 94/93 over two seeds, CelebA 70 vs 79 -- and is still the operating point.

4. Six controls now bound the shape mechanism and all fail: up-ramp, flat, pushing either end alone, and both axes of the (level, compression) parameterization. Each trades memorization for quality along a single frontier; none moves the frontier itself. The robust gap is between shape *classes* -- compressed and tilted sit 7-10 FID below uniform and flat at every matched distance, in two independent parameterizations.

Where this leaves us (3/3)

5. Step count is the first knob that leaves that frontier. Every run until today took 20 steps, because the sampler takes one per window entry. Resampling the champion schedule onto 60 steps -- same window at every noise level, finer integration -- gives CIFAR-10 FID 83.3 -> 78.8 and IS 6.44 -> 6.75 while the median distance to the training set *grows*. Caveat: IS is non-monotone, +0.31 is one split-spread, and the copy tail rises 0.2% -> 1.6%. A seed replicate is running; FID is the solid part.

6. Next: re-tune the window schedule at the better step count, then stop fitting shapes. The ImageNet blocker is not the candidate search, which is already O(1) in image size, but the full pass over the training set at every step -- and more steps multiply exactly that cost.